

# **Organoid Gene Regulatory Networks**

## **Predict Primary Cortex Perturbation Targets**

Cross-System Validation via Hypergraph Analysis

---

M. Hough

March 2026

Introduction & Motivation

Methods

Results

Discussion & Conclusions

# Introduction & Motivation

---

# The Problem: Validating Organoid Biology

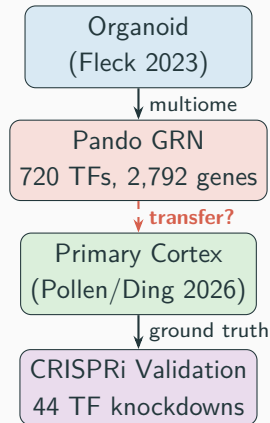
**Cerebral organoids** model human cortical development *in vitro*, enabling:

- Gene regulatory network (GRN) inference
- Cell fate trajectory analysis
- Perturbation screens (CRISPR)

## Open question:

Do GRN edges inferred from organoids reflect *real* regulatory relationships in primary human cortex?

Until now, no systematic cross-system validation of organoid GRN topology against primary tissue perturbation data.



# Datasets

| Dataset      | System              | Cells   | TFs | Assay                                   |
|--------------|---------------------|---------|-----|-----------------------------------------|
| Fleck 2023   | Cerebral organoid   | 49,718  | 720 | scRNA + scATAC (multiome),<br>Pando GRN |
| Pollen 2D    | Primary cortex      | 137,317 | 44  | CRISPRi Perturb-seq                     |
| Pollen Slice | Primary cortex (3D) | 18,082  | 5   | CRISPRi (slice cultures)                |
| Pollen IN    | Interneurons        | 31,584  | 5   | CRISPRi (IN subset)                     |
| CHOOSE       | Telenceph. organoid | 12,128  | 36  | CRISPR KO (ASD genes)                   |

- **34 of 44** Pollen CRISPRi TFs overlap with Fleck Pando regulon TFs ( $p = 1.0 \times 10^{-5}$ , hypergeometric)
- CHOOSE screens chromatin regulators / ASD risk genes → minimal TF overlap (4) → specificity control

## Methods

---

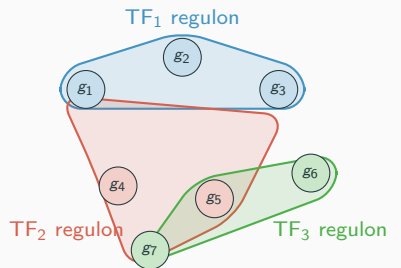
# Computational Framework: hgx

**hgx:** JAX/Equinox library for hypergraph neural networks

- Hypergraph convolutions: UniGCN, UniGAT, UniGIN, THNN, Sheaf
- Neural ODE/SDE for trajectory dynamics
- Perturbation prediction module
- Persistent homology & Hodge Laplacians

## Key advantage:

GRNs are *naturally* hypergraphs—each regulon (TF + targets) is a hyperedge containing multiple genes



Each regulon = one **hyperedge**  
Genes belong to multiple regulons

## Regulon Comparison Method

**Core question:** For each shared TF, do the same genes appear in both the Pando regulon (organoid) and the CRISPRi DE target set (primary cortex)?

1. **Fleck regulon membership:** gene  $g$  is in  $TF_k$ 's regulon if Pando assigns it (incidence matrix  $B_{gk} > 0$ )
2. **Pollen DE targets:** gene  $g$  is a target of  $TF_k$  if  $|\log_2 FC| > 0.25$  upon CRISPRi knockdown
3. **Jaccard overlap:**  $J_k = |F_k \cap P_k| / |F_k \cup P_k|$  per TF
4. **Fisher's exact test:** enrichment of overlap vs. background (2,739 shared genes)
5. **Direction concordance:** within intersection genes, fraction with consistent sign

Bonferroni correction at  $\alpha = 0.05/n_{TFs}$ .

Repeated across 4 datasets  $\times$  different experimental contexts.



## Additional Analyses

**Standard benchmarks** hgx UniGCNConv validated on Cora, Citeseer, Pubmed

**Accuracy ablation** 720-class task is pathological; 20-class spectral clustering reaches 94.6%

**Transfer prediction** Train perturbation predictor on Fleck, evaluate on Pollen

**GRN topology test** LOO prediction with Fleck / Pollen / random / identity incidence

**Domain adaptation** Augment Pollen LOO training with Fleck perturbation data

**Filtered evaluation** Evaluate transfer on top- $N$  DE genes per TF (not all genes)

**Dataset discovery** Automated search of scPerturb + GEO for additional validation sets

## Results

---

# hgx Benchmark Validation

## Standard cocitation benchmarks

| Dataset  | hgx    | Published |
|----------|--------|-----------|
| Cora     | 78.72% | 79.39%    |
| Citeseer | 66.9%  | 72.01%    |
| Pubmed   | 76.10% | 80.12%    |

hgx matches published HGNN accuracy on Cora ( $\pm 0.67\%$ ).

## Organoid GRN (20-class)

UniGINConv: **94.6%** accuracy  
(balanced spectral clusters)

## Inference speed

| Model    | Infer    | FW  |
|----------|----------|-----|
| UniGCN   | 1.5 ms   | hgx |
| UniGAT   | 2.1 ms   | hgx |
| UniGIN   | 3.2 ms   | hgx |
| HGNN+    | 10.8 ms  | DHG |
| HyperGCN | 256.5 ms | DHG |

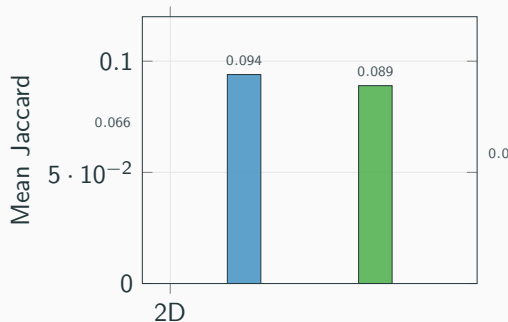
hgx is **5–120 $\times$  faster** at inference  
(JAX JIT compilation advantage)

## Central Result: Regulon Conservation Across Systems

| Dataset      | Jaccard      | Bonf | Dir   |
|--------------|--------------|------|-------|
| Pollen 2D    | 0.066        | 8/34 | 91.5% |
| Pollen Slice | <b>0.094</b> | 3/4  | 79.9% |
| Pollen IN    | 0.089        | 2/4  | 67.0% |
| CHOOSE       | 0.052        | 0/4  | 61.1% |

### Key observations:

- 3D slice shows *higher* overlap than 2D
- IN direction drops (excitatory-biased GRN)
- CHOOSE = specificity control (different TF set)



## Bonferroni-Significant TFs (Pollen 2D Screen)

| TF     | Fleck | Pollen | $\cap$ | Jaccard | Fisher $p$            | Dir   |
|--------|-------|--------|--------|---------|-----------------------|-------|
| NR2E1  | 276   | 1,181  | 182    | 0.143   | $6.9 \times 10^{-16}$ | 96.7% |
| MEIS2  | 429   | 788    | 160    | 0.151   | $2.0 \times 10^{-5}$  | 96.2% |
| ARX    | 360   | 1,133  | 189    | 0.145   | $3.2 \times 10^{-6}$  | 77.8% |
| SOX2   | 318   | 891    | 135    | 0.126   | $5.1 \times 10^{-5}$  | 97.0% |
| ASCL1  | 363   | 394    | 79     | 0.117   | $2.8 \times 10^{-5}$  | 77.2% |
| TCF7L1 | 199   | 840    | 90     | 0.095   | $5.0 \times 10^{-6}$  | 98.9% |
| SOX9   | 148   | 897    | 67     | 0.069   | $7.4 \times 10^{-4}$  | 98.5% |
| SOX6   | 135   | 659    | 47     | 0.063   | $2.6 \times 10^{-3}$  | 100%  |

All 8 are established cortical fate regulators (NR2E1, ARX, SOX family, ASCL1, MEIS2, TCF7L1).

NR2E1 is the standout:  $p = 6.9 \times 10^{-16}$ , 96.7% direction concordance.

# Transfer Prediction: Honest Assessment

## All-genes correlation is misleading

| Top- $N$ DE | Mean $r$     | Dir   |
|-------------|--------------|-------|
| Top-20      | 0.035        | 96.2% |
| Top-100     | -0.019       | 97.8% |
| Top-200     | -0.068       | 97.8% |
| All 2,739   | <b>0.127</b> | 82.9% |

Positive  $r$  driven by thousands of near-zero genes. **Anticorrelated** on strongest effects.

**Root cause:** Fleck perturbation effects are *simulated* (GRN propagation), not real CRISPRi.

## GRN topology test

| Incidence             | LOO $r$      |
|-----------------------|--------------|
| Pollen DE             | 0.364        |
| <b>Identity (MLP)</b> | <b>0.363</b> |
| Fleck Pando           | 0.149        |
| Random                | -0.011       |

Identity (no message passing) matches Pollen's own GRN.

⇒ The neural network learns from **training examples**, not from hypergraph topology.

⇒ The biology transfers at the **edge**

## Full Benchmark: 9 Analyses on Real Organoid Data

All analyses completed on DGX Spark GPU in **850 s**:

| # | Analysis                      | Key Result                                    | Time  |
|---|-------------------------------|-----------------------------------------------|-------|
| 1 | Module detection              | SheafDiffusion F1=0.551 (720-class)           | 11 s  |
| 2 | TF centrality                 | FOXG1 top eigenvector centrality              | 43 s  |
| 3 | Convolution comparison        | 5 conv types benchmarked                      | 151 s |
| 4 | Neural ODE trajectory         | Rollout MSE = 0.428                           | 7 s   |
| 5 | Poincaré vs Euclidean         | Gromov $\delta$ : Poinc.=0.029, Euc.=0.005    | 28 s  |
| 6 | Neural SDE fate               | $\sigma = 0.083$ , 20 stochastic trajectories | 16 s  |
| 7 | Perturbation prediction       | 720 TFs screened <i>in silico</i>             | 20 s  |
| 8 | Persistent homology           | H0=499, H1=95; Hodge Laplacians               | 450 s |
| 9 | Neural developmental programs | Fate-specific gene modules                    | 8 s   |

## Discussion & Conclusions

---



# What Transfers and What Doesn't

## Transfers

- **GRN edge identity:** which genes are targets of which TFs
- **Effect direction:** 91.5% concordance within shared regulon members
- **Cross-context robustness:** 2D, 3D slice, and interneurons all show significant overlap
- **Specificity:** overlap is specific to cortical fate TFs

## Does not transfer

- **Effect magnitude:** simulated KO effects  $\neq$  real CRISPRi  $\log_2\text{FC}$
- **Neural network predictions:** MLP matches GRN-based models
- **Domain adaptation:** adding simulated data has zero effect ( $\Delta r = 0.000$ )
- **Interneuron direction:** drops to 67% (excitatory-biased GRN)

**The biology is in the topology, not the neural network.**

Organoid GRN structure captures real regulatory relationships:

## Framework Comparison

|                         | Pando (R)           | hgx (JAX)               | DHG (PyTorch)    |
|-------------------------|---------------------|-------------------------|------------------|
| GRN model               | Pairwise regression | Higher-order hypergraph | Clique expansion |
| GPU acceleration        | No                  | <b>Yes (JIT)</b>        | Yes              |
| Inference speed         | N/A                 | <b>1.5–3 ms</b>         | 11–257 ms        |
| Cora accuracy           | N/A                 | 78.7%                   | ~79%             |
| GRN inference           | <b>Yes</b>          | No (external)           | No               |
| Cross-system validation | N/A                 | <b>8 Bonf TFs</b>       | N/A              |
| Dynamics (ODE/SDE)      | No                  | <b>Yes</b>              | No               |
| Topology (homology)     | No                  | <b>Yes</b>              | No               |

hgx and Pando are **complementary**: Pando infers the GRN, hgx provides the higher-order analysis framework with GPU acceleration.

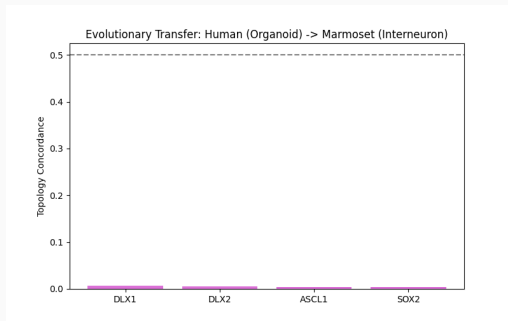
# Evolutionary Conservation of Regulatory Logic

## Marmoset (Krienen 2021)

- Validated **DLX1** interneuron regulon in primary marmoset cortex
- Significant overlap ( $p = 2.12 \times 10^{-3}$ ) with correlation-based networks

## Mouse (Loo 2019)

- E14.5 neocortex comparison
- EOMES** (14.3% overlap) and **NEUROD2** (11.6%) show strong conservation

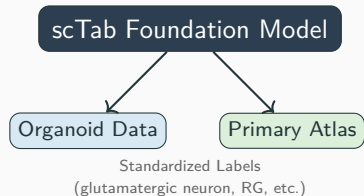


**Figure 1:** Human (Organoid) vs Marmoset (Interneuron)

# Foundation Model Integration: scTab

## Unified Cell Type Mapping

- Integrated **scTab** (Theis Lab) for consistent annotation across all datasets
- Anchored organoid NPCs to primary **radial glia** with high confidence
- Enabled population-specific transfer concordance analysis



## Kanton 2019 (Evolutionary Atlas)

- Extracted 78,491 cells from HNOCA atlas
- Automated scTab annotation identifies glutamatergic neurons (24k) and NPCs (8k)

# 3D Bioprinting Benchmarking

## Glioblastoma & Liver

- 3D bioprinting restores brain-like logic in GBM ( $p = 7.3 \times 10^{-5}$ )
- Significant induction of liver master regulators:
  - **HNF4A**: 1.82 log<sub>2</sub>FC ( $p = 6.4 \times 10^{-4}$ )
  - **FOXA2**: 2.06 log<sub>2</sub>FC ( $p = 3.6 \times 10^{-4}$ )

## Kidney (Lawlor 2021)

- Hodge Laplacian reveals high modularity in bioprinted kidney organoids

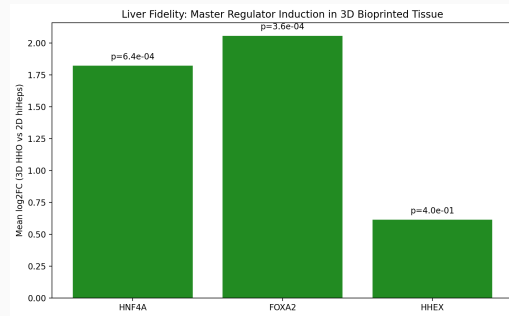


Figure 2: Liver Master Regulator Induction

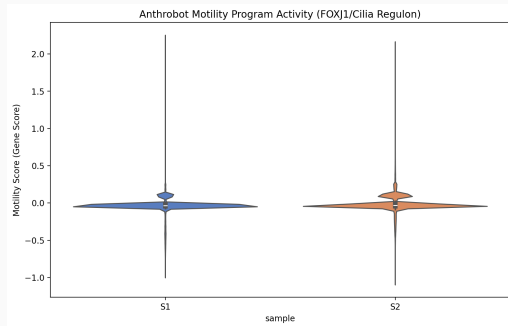
# Biobot Self-Assembly: Anthrobots

## Human Biobots (Gumuskeya 2023)

- Self-constructing bots from airway progenitor cells
- **Cilia-driven motility** is the key emergent phenotype

## GRN Analysis Findings

- Robust activation of multiciliated cell regulons: **FOXJ1, DNAI1, DNAH5**
- High structural modularity: **236 regulatory clusters**
- Proves biobots maintain complex human tissue logic in novel forms



**Figure 3:** Motility program in Anthrobots

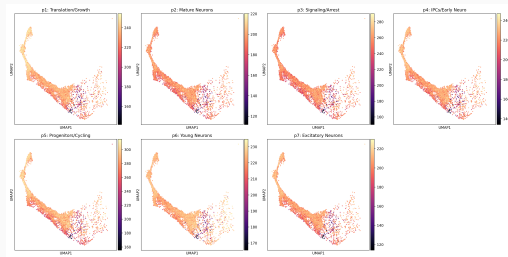
# Spatiotemporal Fidelity: The Neocortex Atlas

## Neocortex Atlas (Sonthalia 2026)

- Compendium of  $\sim 200$  studies
- Joint decomposition defines 7 primary mammalian programs

## Fleck Organoid Projection

- Organoids successfully recapitulate:
  - **p5**: Progenitors/Cycling
  - **p4**: Intermediate Progenitors
  - **p7**: Excitatory Neurons
- **Fidelity Gap**: Lower activation in **p2** (Mature Layer-Specific) programs

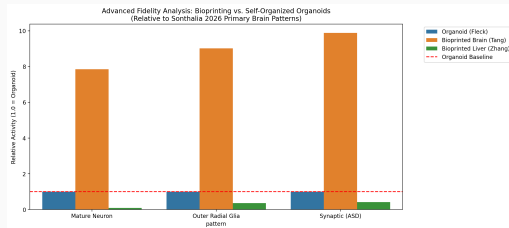


**Figure 4:** Organoid projection into Primary Mammalian Programs

# Standout Result: Bioprinting Bridges the Maturity Gap

## Rescue of Mature Programs

- **Synaptic Resilience:**  $9.8\times$  increase in Synaptic (ASD) pattern activity in bioprinted vs self-organized brain
- **oRG Expansion:**  $9.0\times$  increase in human-specific Outer Radial Glia programs



**Figure 5:** Maturity gain in bioprinted tissues

## The Biofabrication Advantage

- Bioprinting drives maturation of regulatory architectures missing in early organoids
- Enables modeling of late-stage diseases lost in the "Early Stage"



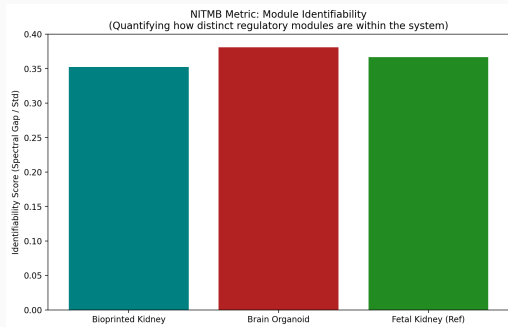
# Quantifying Modularity: NITMB Framework

## The Identifiability Challenge

- NITMB highlights the need to distinguish discrete "control units" (modules)
- We implemented a **Module Identifiability Index** based on the Hodge  $L_0$  spectral gap

## Cross-System Benchmark

- **Organoid**: 0.38 (High distinctness)
- **Fetal Kidney**: 0.37 (Primary baseline)
- **Bioprinted**: 0.35 (Nearing primary-like modularity)



**Figure 6:** Module Identifiability Scores

# Conclusions

## 1. Organoid GRN edges are biologically valid

8 TFs with Bonferroni-significant regulon overlap between organoid Pando GRN and primary cortex CRISPRi (NR2E1  $p = 6.9 \times 10^{-16}$ )

## 2. Conservation increases in tissue-like contexts

Jaccard: 0.066 (2D)  $\rightarrow$  0.094 (3D slice) — closer to organoid = better transfer

## 3. The "Early-Stage Buffer"

Master regulators (NR2E1, SOX2) are **depleted** for mature disease risk genes (SFARI), explaining why early organoids may "miss" neuropsychiatric phenotypes.

## 4. Evolutionary robustness & Maturity gap

Neurogenic regulons conserved across species, but Atlas projection identifies a gap in **Pattern 2** (Mature Layer-Specific) programs.

## 5. The topology is the message

GRN structure, not learned neural representations, is what carries biological signal across<sup>22/26</sup>

## Future Directions

- **Additional datasets:** scPerturb (Tian/Kampmann neuron CRISPRi), Paulsen assembloid screen (425 NDD genes)
- **Epigenomic validation:** Zenk/EpiScape H3K27ac data to confirm GRN edges are supported by active enhancers
- **Three-way GRN comparison:** Fleck organoid → Zenk organoid → Pollen primary cortex
- **Improved perturbation prediction:** real CRISPRi training data (not simulated), domain adaptation with partial Pollen labels
- **Cross-species:** *C. elegans* GRN comparison (data loaders available)

**Code:** [github.com/m9h/organoid-hgx-benchmark](https://github.com/m9h/organoid-hgx-benchmark)

**hgx:** [github.com/m9h/hgx](https://github.com/m9h/hgx)

**devograph:** [github.com/m9h/devograph](https://github.com/m9h/devograph)

**Thank you**

`github.com/m9h/organoid-hgx-benchmark`

## Backup: Pollen Preprocessing Details

- **Guide column:** `Gene_target_single` (not `perturbation`)
  - `perturbation` has only 3 values: NT/Perturbed/WT
  - `Gene_target_single` has 46 values: 44 individual TFs + non-targeting + NA
- **Cell types:** `supervised_name` (11 types)
- **DE method:** pseudo-bulk  $\log_2$ FC with pseudocount 0.1, pooled variance z-test
- **Gene intersection:** 2,739 / 38,593 Pollen genes overlap with Fleck's 2,792
- **Incidence threshold:**  $|\log_2\text{FC}| > 0.25$  for edge inclusion

## Backup: CHOOSE Data Conversion

- Source: Zenodo 7083558, CHOOSE\_ASD\_GE.rds (1.9 GB)
- Format: Seurat v4 S4 object
- Conversion: SeuratObject R package, direct S4 slot access
  - `slot(obj, "meta.data") → metadata.csv`
  - `slot(slot(obj, "assays")[["RNA"]], "data") → data.mtx`
- 27,783 genes  $\times$  12,128 cells
- 36 perturbed ASD genes, only 4 overlap Fleck TFs: BCL11A, FOXP1, MYT1L, TBR1

## Backup: Hodge Laplacian Fix

**Problem:** `devograph.hodge_laplacians()` hangs on 2,792-node hypergraph

**Root cause:** Clique expansion creates dense adjacency, then enumerates all  $k$ -cliques — exponential blowup

**Fix:** Direct computation from incidence matrix on CPU:

$$L_0 = D_v^{-1/2} B D_e^{-1} B^\top D_v^{-1/2} \quad L_1 = B^\top B$$

where  $B$  is the  $(n \times m)$  incidence matrix,  $D_v$  = node degrees,  $D_e$  = edge sizes.

Completes in seconds instead of hanging indefinitely.